



Combined treatment of alkaline and disperser for improving solubilization and anaerobic biodegradability of dairy waste activated sludge

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HIGHLIGHTS

- Chemo-mechanical pretreatment synergistically enhances sludge solubilization.
- No study carried out on this pretreatment method.
- The optimal study conditions were, SE of 4544 kJ kg⁻¹ TS for NaOH – pH 10.
- COD solubilization and SS reduction was 24% and 23.3% higher than control.
- Biogas production was 76% higher than control, at optimized conditions.

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ABSTRACT

An investigation into the influence of combined alkaline and disperser pretreatment on sludge disintegration was studied. The effects of four variables, alkalines (NaOH, KOH, Ca(OH)₂), treatment time (15–180 min), pH (8–11) and rpm (4000–24,000) were investigated. The effect of sludge pretreatment was evaluated by COD solubilization, suspended solids reduction and biogas production. The best performances, in terms of COD solubilization, SS reduction and biogas production, were the ones that occurred for specific energy input of 4544 kJ kg⁻¹ TS for NaOH at pH10, were found to be 24%, 23.3% and 76%, higher than the control, respectively. Not only the increase in biogas production was investigated, excluding protein hydrolysis was also performed successfully by this combined pretreatment even at low specific energy input. Thus, this chemo-mechanical is an effective method for enhancement of biodegradability and it laid the basis to produce higher biogas quantities, to improve clean energy generation from WAS.

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1. Introduction

Dairy industry is one of the prime sectors of an Indian industrial realm. Water management in the dairy industry is well documented, however effluent production and disposal remain a problematical issue for the dairy industry. The expected increase is itself a challenge for waste management and the choices of treatment and disposal methods will have large economic and environmental implications. Sludge management places a high economic burden as it occupies approximately 60% of the plant's operating charges. Because of the high cost of these treatments, interest for solutions allowing sludge volume and mass reduction is escalating. Anaerobic digestion is considered as the sustainable option in treatment of solid wastes, can be explained considering mainly three factors (1) the need to apply a process to dispose of organic

solid wastes more environmental friendly than landfills as requested by the latest rules concerning the environmental protection in many countries in the world; (2) the opportunity to obtain from this process a renewable fuel called biogas, an alternative to fossil fuels; (3) the advantage of relatively low costs in starting up and managing this process (Esposito et al., 2012). Historically, the anaerobic digestion process applied to sludge treatment is critically considered in favor of the sludge stabilization to reduce odors and pathogens. Currently, emphasis is captured on exploiting and utilizing its actual and potential ability for energy conservation and recovery. Anaerobic degradation can be achieved through several stages: hydrolysis, acidogenesis, acetogenesis and methanogenesis. The efficiency of digestion has often been limited by hydrolysis, which is widely regarded as the rate limiting step due to the complex structure of Waste Activated Sludge (WAS). Consequently, longer sludge retention times (SRTs) are required for digestion of WAS. In order to improve the rate of hydrolysis and the anaerobic digestion performance, sludge

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disintegration was developed as a pretreatment process to accelerate the anaerobic digestion and to increase the degree of stabilization. The advantage of sludge disintegration is 2-fold. First, the amount of released soluble substrate significantly enhances volatile fatty acids (VFA) generation for improved gas production during anaerobic digestion. Secondly, pretreatment decreases the viscosity of the sludge, permitting greater solids concentration to be fed to an anaerobic digester (Saha et al., 2011). Various methods like alkaline treatment (Kim et al., 2003), microwave radiation (Ahn et al., 2009), thermo-alkaline (Uma Rani et al., 2012), Ultrasonic treatment (Zhang et al., 2007) and biological hydrolysis with enzymes (Gopi Kumar et al., 2012) were investigated for sludge disintegration by several researchers in half-scale and lab-scale plants to bypass the rate limiting stage of hydrolysis.

Among these alternate technologies, mechanical treatment of sludge plays an important role because it favors solubilization of particulate matters in liquid phase. Once the organics are solubilized, the sludge becomes more convenient for further treatment. The mechanical treatment methods which are used as pretreatment techniques include mainly ultrasonication and homogenization (Dispersion). The principle of ultrasonic treatment relies on the induced cavitation process. Through subsequent compression and expansion of the fluid under the effect of the ultrasonic waves, implosions are generated, which give rise to local extreme conditions. Because of these, $\text{OH}^\cdot$, $\text{HO}_2^\cdot$, $\text{H}^\cdot$ radicals and hydrogen peroxide can be formed. Thus, the different phenomena: oxidizing of free radicals, pyrolysis, combustion and shearing are responsible for the sludge disintegration. Although many researchers have shown its effectiveness, power consumption and the frequent replacement of the ultrasound probes in every 1.5–2 years impedes its practical application (Zhang et al., 2007). In comparison to this, dispersion is also a well known mechanical method for cell disruption. Dispersion is the dissolution and diffusion of a solid, liquid or gaseous phase in a continuum that is not consolute with that phase. Of late, this has been reported as a sludge pretreatment method, which changes both the rate and extent of sewage sludge degradation in the anaerobic digestion process (Rai and Rao, 2009). It was mainly developed for stabilization of food and dairy emulsions, and has opened up more and more new application areas (Jacquel et al., 2008). This is generally based on the disruption of microbial cell wall and the release of cytoplasm into liquid phase by functions of pressure gradient, cavitation, turbulence, impingement, shear stresses and extensional shear (Zhang et al., 2008). It has several advantages in comparison to other sludge disintegration methods, i.e. no chemical changes or denaturing during cell lysis, easy operation, and high lysis efficiency. The high-energy requirement in dispersion treatment can be reduced by combining with other treatment methods based on that different mechanisms of action could allow obtaining higher disintegration. In comparison with others, alkaline pretreatment is a commonly examined method whose mechanism is to induce swelling of particulate organics at high pH, making the cellular substances more susceptible to enzymatic action and also it has advantages of simple device, convenient operation and high efficiency (Kim et al., 2003). The literature on sludge dispersion is not so extensive, but full-scale applications combined with alkaline treatment exist. This combined method was trialed at the Chilliwack WWTP and resulted in improved VS reduction in mesophilic digesters by 18% points from 60% to 78% (Rabinowitz and Stephenson, 2005).

Consequently, this study uses alkaline pretreatment to weaken the cell walls, as well as a disperser for a sudden pressure change to lyse the cells of dairy waste activated sludge before anaerobic digestion and it is also observed that works concerning the optimization of chemo (alkaline)-mechanical (dispersion) disintegration of dairy waste activated sludge at low specific energy and short duration to improve biogas yields are not readily available. The

influence of different pretreatment conditions was evaluated in terms of (a) sludge solubilization, (b) suspended solids reduction, and (c) biogas production.

2. Methods

2.1. Sludge sampling and characterization

Dairy waste activated sludge was sampled from Aavin dairy effluent treatment plant at Ambathur, Chennai. Samples were collected and stored at 4 °C. The characteristics of the raw sludge were as follows: total COD (TCOD) was 10200 mg/L, soluble COD (SCOD) was 600 mg/L, pH was 6.97, total solids (TS) concentration was 13850 mg/L, suspended solids (SS) concentration was 9900 mg/L and volatile solids concentration was 7250 mg/L.

2.2. Chemo-mechanical pretreatment

This study was proposed to develop a new pretreatment approach to disintegrate the sludge. In this chemo-mechanical pretreatment, without temperature regulation, the sludge pretreatment was performed as batch tests with a volume of 500 mL sludge sample. For the waste activated sludge studied, different alkalines (NaOH , KOH , Ca(OH)_2) of different concentrations were added to reach different pH values of 8, 9, 10 and 11. Then, sludge disintegration were carried out using a disperser (IKA T25 Ultra Turrax Digital disperser), which is a high-speed shearing device to disperse large solids into small particles. Due to the high rotation speed of the rotor, the medium to be processed is automatically drawn axially into the dispersion head and then forced radially through the slots in the rotor/stator arrangement. The high accelerations acting on the material produce extremely strong shear and thrust forces. In addition, high turbulence occurs in the shear gap between rotor and stator, which provides optimum mixing of the suspension. The dispersion effectiveness is heavily dependent on the product of the shear gradient and the time the particles spend in the shear zone (IKA Service Manual, 2007). A series of experiments were performed at different working rpm range of 4000–24,000 in order to enhance solubilization of particulate material, as well as enzymatic hydrolysis. Moreover, the effect of this pretreatment mainly depends on treatment time, and it was evaluated by taking samples at different times (15–180 min) in order to study the combined effect.

2.3. Biochemical methane potential (BMP) assay

In recent years, interest in biochemical potential (BMP) tests has increased as reflected by the wide range of research papers dealing with the BMP assays. The BMP assay can be used as an index of the anaerobic biodegradation potential, and thus, relative residence times required for complete digestion (Labatut et al., 2011). The BMP is measured with the BMP test, which consists in measuring the bio-methane or biogas produced by a known quantity of waste in batch and anaerobic conditions. Biogas production of raw and pretreated sludge samples was initially determined by batch tests at mesophilic conditions. The biodegradability assays were conducted in anaerobic batch reactors of 300 mL capacity serum bottles with hermetically sealed stoppers. Each serum bottle was filled with 100 mL of inoculum and 50 mL of substrate. A blank treatment with only 150 mL of inoculum was used to determine biogas production due to endogenous respiration. The inoculums needed to be quite active, with a good adaptability and a low endogenous respiration (Bougrier et al., 2005). For these experiments, the rumen micro-organisms of cattle dung were used as an inoculum. Moreover, in case of anaerobic biodegradability, the

use of a highly active animal inoculum waste will reduce the experimental time significantly, or reduced the amount of inoculum required in full-scale batch digesters, and consequently, the corresponding digester volume (Borja et al., 2003). The initial pH was recorded as neutral before the start of BMP assay tests. After adding the substrates and inoculum, the reactors were closed with a rubber septum and an aluminum seal to make them air tight and was subsequently purged with nitrogen gas at the rate of 10 mL per second for 20 min into the reactors to maintain anaerobic conditions. An internal temperature of 35 °C was maintained by incubating the reactors in a temperature controlled mechanical shaker (220 rpm). Batch reactors were operated with a residence time of 42 days. Samples were analyzed in duplicate. Enhancement of biodegradability was evaluated by comparison of biogas volumes produced by treated and untreated samples. Based on the outcomes of the preliminary investigations, the conditions of the batch anaerobic reactors were selected. The anaerobic biodegradability test's performance was evaluated by fitting the cumulative biogas production data to the modified Gompertz equation. The Gompertz equation describes cumulative biogas production assuming that, biogas production is a function of bacterial growth (Redzwan and Banks, 2004). The modified Gompertz equation is presented below as Eq. (1):

$$B = P\{1 - \exp[-R_m(t - \lambda)/P]\} \quad (1)$$

where B is the cumulative biogas production (mL), R_m is the maximum biogas production rate, P is the biogas yield potential (mL), and λ is the duration of lag phase, days. Using Matlab 7.0, the unknown parameters were calculated in order to measure the difference between the experimental measurement and the corresponding stimulated value.

2.4. Analytical parameters

The following parameters were analyzed before and after thermo-chemical treatment: total solids (TS), suspended solids (SS), chemical oxygen demand (COD), carbohydrate concentration, protein concentration and pH as per Standard Methods for the Examination of Water and Wastewater (American Public Health Association, 2005). The analyses were performed on both the sludge and the supernatant to identify the total and soluble fractions of the specific component. TS, VS and SS were measured in the whole sludge (Total TS, VS and SS) and in the supernatant after centrifugation at 30,130g, for 15 min and subsequent filtration through 0.45 µm microfiber filter paper. The methane content in the biogas was analyzed using a Baroda gas chromatograph (GC) equipped with a thermal conductivity detector and porapack Q column with hydrogen as carrier gas at a flow rate of 40 mL/min. Microscopic assessments were analyzed with Zeiss micro imaging microscope. Microbial populations present in the raw and pretreated sample (at optimized condition) were enumerated by spread plate technique in nutrient agar. A dilution series (10^{-2} to 10^{-4}) of each sample was prepared by serially transferring 1 mL of sludge. Plates were incubated at 37 °C for 24 h (American Public Health Association, 2005). The microbial populations obtained in both samples were enumerated and compared.

Protein concentration was determined on total sludge and on the supernatant using the Lowry method (Takahashi et al., 2009). Following reactions with salts and Folin's phenol reagent, absorbance of samples was determined at 620 nm, using a spectrophotometer. Carbohydrate concentration was determined on total sludge and on the supernatant using the anthrone method (Tapia et al., 2009). Subsequent to reactions with anthrone and sulfuric acid, absorbance of samples was determined at 625 nm, using a spectrophotometer. The COD solubilization (%) was calculated by using the Eq. (2),

$$\text{COD solubilization (\%)} = \frac{\text{Soluble COD(SCOD)measured after pretreatment}}{\text{Total COD(TCOD)measured after pretreatment}} \times 100 \quad (2)$$

2.5. Specific mechanical energy (SE)

As it is well known that the treatment time and the applied disperser rpm are the major parameters influencing the process, we have defined an energy parameter called the specific energy, the energy input related to treated solid mass. It was considered as a main variable parameter for evaluation of disintegration performance of the sludge. The specific energy consumed for the treated solid mass was calculated by using Eq. (3),

$$\text{SE}(\text{kJ kg}^{-1} \text{ TS}) = \text{Power/Mass of Total Solids} \quad (3)$$

where power is presented as a function of angular velocity (ω) and torque (τ). The angular velocity ultimately relates the rpm to the specific energy, since rpm is one of the major variable parameter in the pretreatment conditions. Whereas torque experienced by the disperser rotor relates the load imposed to the specific energy.

2.6. Data analysis

Methane gas production was calculated from the total volume of biogas produced and, headspace measurements of gas composition at each time interval, using the mass balance Eq. (4),

$$V_M = \sum_{i=1}^{i=n} (G_i C_i) + V_h C_n \quad (4)$$

where V_M is the cumulative volume (mL) of methane produced at t_n , G_i is the volume (mL) of gas produced at any incubation time t_i , between the first measurement ($i = 1$) and t_n , C_i is the methane proportion in the gas sample collected at any incubation time t_i , between the first measurement ($i = 1$) and t_n , C_n is the proportion of methane in the gas sample collected at t_n , and V_h is the volume (mL) of headspace in the serum bottle (Lopez et al., 2007).

In the BMP test, the COD to methane conversion was calculated by using Eq. (5),

$$\text{COD consumption (g/l)} = (V_{gs} - V_{gk})(\% \text{CH}_4)/V_s(395) \quad (5)$$

The converted COD efficiency can be calculated by using Eq. (6),

$$\text{Converted COD efficiency} = \frac{\text{COD consumed (g/l)}}{\text{Total COD (g/l)}} \times 100 \quad (6)$$

where V_{gs} , the gas accumulation produced by the sample (mL), V_{gk} is the gas volume accumulation produced by control (mL), V_s is the sample volume added to the test bottle (mL), % CH_4 , the CH_4 composition in the gas by GC, the 395 value was the volume of CH_4 produced for every g of COD consumed at 35 °C, 1 atm (Desiana and Setiadi, 2009).

3. Results and discussion

3.1. Chemo-mechanical pretreatment

In this study, chemo-mechanical pretreatment of waste activated sludge was performed to improve the treatment efficiency. Albeit it is a user-friendly technique and shortens the conventional long retention times, as it requires high energy usage, the operating parameters need to be optimized in order to get cost-efficient results. The control was performed using non-pretreated WAS. The treatment was performed for different rpm, alkalines, pHs and treatment time. The expected effect of this combined

treatment of sludge was an increase in soluble materials, with interest focused on COD solubilization, suspended solids reduction and biogas production, consequently enhancing hydrolysis.

3.1.1. COD solubilization and SCOD release

The pretreatment was done to improve the bioavailability of sludge particulate material. SCOD calculations were considered the main parameter for evaluation of sludge particulate material, and it enables an evaluation of the maximum level of sludge solubilization. Increased SCOD is determined as the substance that can be readily used to produce methane during anaerobic digestion (Wang et al., 2005). Fig. 1 demonstrates the effects of different alkali agents on COD solubilization during the chemo-mechanical pretreatment. Fig. 1a demonstrates the effect of NaOH on COD solubilization. COD solubilization was analyzed at different pH conditions of 8–11 for various time intervals from 15 to 180 min and at various rpm ranging from 4000 to 24,000. As estimated, all pretreatment conditions caused significant increases, when compared to control. From the figure, it is evident that, as the treatment time was increased from 15 to 180 min, an increase in COD solubilization was observed. This increase can be explained by the hydrolysis of the large organic molecules, the lysis of the cell walls and the

disintegration of the sludge flocs, which was intensified by the applied chemo-mechanical treatment. As a result intracellular organic material is released to the liquid phase and increases the SCOD. Nonetheless, irrespective of time, the rpm plays a major role in enhancing COD solubilization. The COD solubilization % increased significantly with the increase of rpm from 4000 to 24,000. For example, at pH 8, the increase in rpm from 4000 to 24,000 rpm increases COD solubilization from 16% to 21.8%. The other pH conditions with an increase in rpm also follow the similar trend in COD solubilization. These results demonstrated that larger quantities of sludge organic matters were released from solid phase into liquid phase at higher rpm. Higher rpm benefit the sludge disintegration because of thermal effect in addition to pressure gradient effect. With the increase of rpm from 4000 to 24,000, it is clear that rpm below 8000 doesn't help in solubilizing the excess sludge, and an effective rise is visualized only after 12000 rpm. Similarly, the COD solubilization increased with the increase of pH, which was not consistent with the previous reports that the degradation rate decreased at high pH values (He et al., 2007). For this reason, the sludge became basic with the increase of pH, and the alkaline destroyed floc structures and cell walls by hydroxyl anions. However, pH 8 and 9 seem to be too low for an effective COD

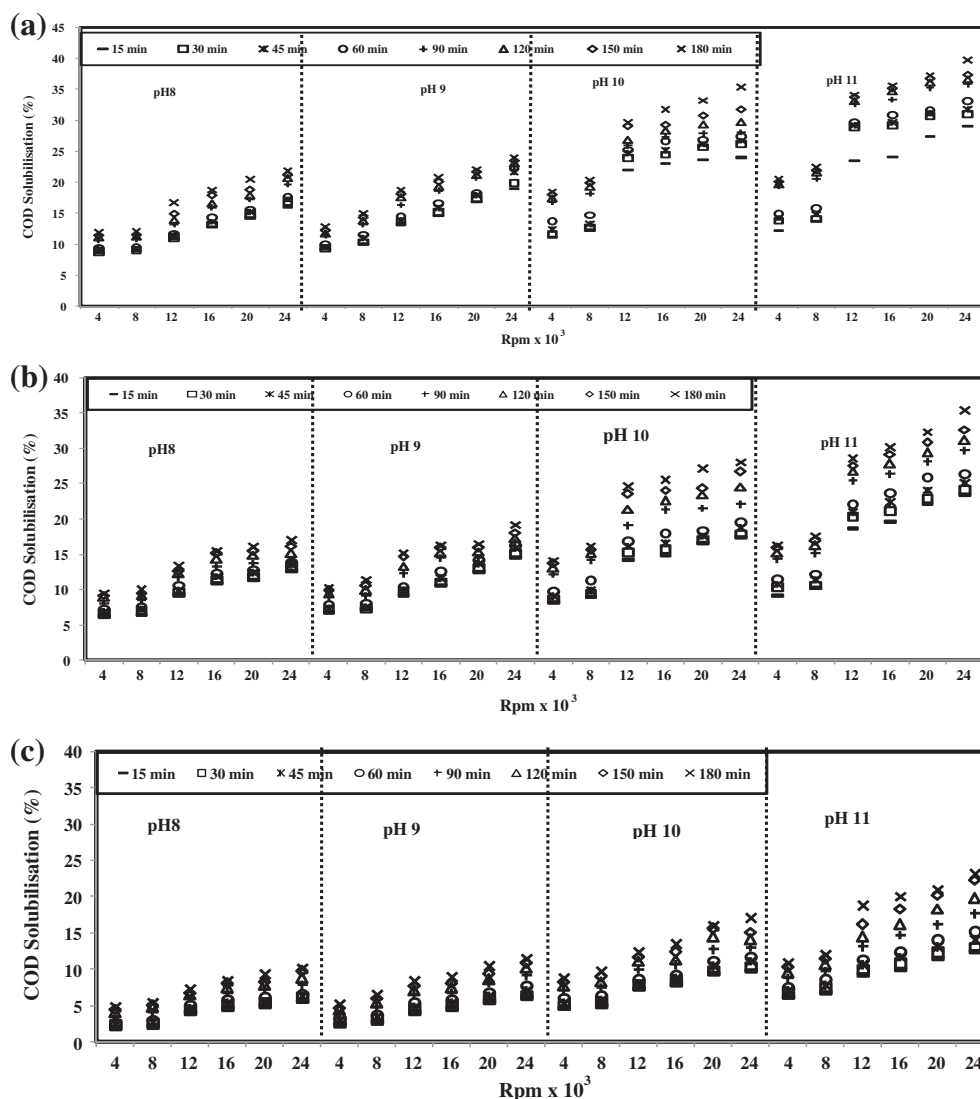


Fig. 1. Effects of different alkali agents on COD solubilization. (a) Effect of NaOH on COD solubilization. (b) Effect of KOH on COD solubilization. (c) Effect of Ca(OH)₂ on COD solubilization.

solubilization, in view of the fact that even after a treatment time of 180 min only a very limited amount was set free. At this stage of experimentation due to less effectiveness, the pH conditions of 8 and 9 and rpm of 4000 and 8000 are neglected.

Fig. 1b demonstrates the effect of KOH on COD solubilization. Similar results were obtained when the experiments were carried out with KOH. As seen from the figure, the COD solubilization % was less in the case of KOH compared to NaOH, but the variation in the trend with respect to pH and rpm were similar. While at pH conditions of 8, 9, 10 and 11 at rpm range of 4000–24,000 rpm for 180 min, the solubilization was found to be 12.6–17%, 14.6–19.2%, 17.4–28.1% and 23.4–35.5%, respectively. Considering COD solubilization, out of comparison it could be distinguished that the COD solubilization at pH 10 using NaOH yielded similar results as in the case with KOH at pH 11. However, the consumption of KOH to obtain the respective pH was higher than that of NaOH. For example, to achieve pH 10, a NaOH dosage of 0.007 g/g SS was required, whereas to achieve pH 10, KOH demands an alkali dosage of 0.03 g/g SS. Fig. 1c demonstrates the effect of $\text{Ca}(\text{OH})_2$ on COD solubilization. From the figure it could be distinguished that, the COD solubilization% was very much reduced in the case of $\text{Ca}(\text{OH})_2$, but the variation in the trend with respect to pH was similar. For example, at various pH conditions of

8 to 11 and at rpm range of 4000–24,000 rpm for 180 min, the solubilization was found to be 5.7–10.2%, 6.2–11.5%, 9.9–17.2% and 12.4–23.2%, respectively. The effect of rpm could also be evaluated from the study. In all the alkaline conditions, the rise in COD solubilization is very low in the rpm conditions of 4000 and 8000 and at pH conditions of 8 and 9.

3.1.2. SS Reduction

SS reduction is an indication of sludge stability, and it is used for assessing the effectiveness of a process in stabilizing sludge and reduction (Gholamreza et al., 2008). Fig. 2 shows the effects of different alkali agents on SS reduction during the pretreatment. The reduction in the SS concentration was observed during all the pretreatment conditions. Fig. 2a shows the effect of NaOH on SS reduction. As seen in figure, when the pH, rpm and treatment time was increased, increase in SS reduction was observed. For example, at pH 8, the increase in rpm from 4000 to 24,000 rpm increases SS reduction from 15 to 20.3%. The other pH conditions with an increase in rpm also follow the similar trend in SS reduction. This proves that chemo-mechanical treatment at lower rpm had a slight effect on biodegradability, in addition to the higher rpm with enhanced irradiation time seemed to be more efficient. Thus, sludge solid structure was destroyed and organic components dissolve

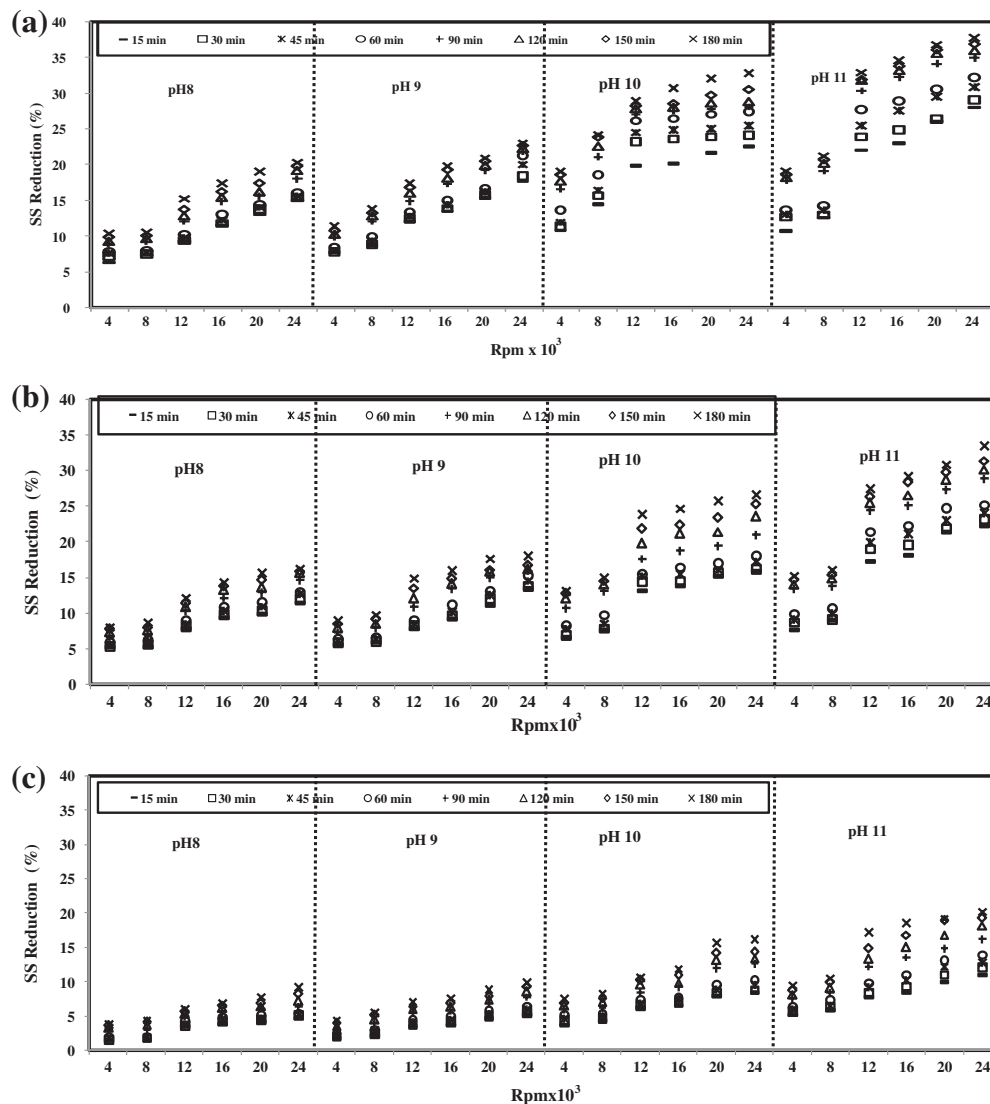


Fig. 2. Effects of different alkali agents on SS reduction. (a) Effect of NaOH on SS reduction. (b) Effect of KOH on SS reduction. (c) Effect of $\text{Ca}(\text{OH})_2$ on SS reduction.

into soluble parts. As a result, the solid content of the sludge decreased and the amount of sludge reduced.

Fig. 2b and c demonstrates the effect of KOH and Ca(OH)_2 on SS reduction. Similar results were obtained when the experiments were carried out with KOH and Ca(OH)_2 . The figure clearly is with consistent that for KOH and Ca(OH)_2 , the increases in SS reductions were comparatively lower than NaOH. A higher solubilization degree of suspended solids in sludge is important for elimination of the hydrolysis phase of anaerobic biodegradation. Hence, suspended solids solubilization was proportionally more significant than total solids solubilization for sludge disintegration and the SS reduction efficiency was found to be slightly lower than the COD solubilization efficiency. Like COD solubilization, increase in SS reduction is very low in the rpm conditions of 4000 and 8000 and at pH conditions of 8 and 9.

Thus, a wider range of optimal values were selected based on COD solubilization and SS reduction. Further studies were carried out above 12,000 rpm and at pH conditions of 10 and 11 in all the alkaline used. The final optimal parameters are obtained based upon the specific energy input for the corresponding rpm and time.

3.1.3. Effect of specific energy on COD solubilization

In this study, the COD solubilization was correlated with the specific energy input. Such correlations will help to optimize the energy needs to achieve maximum sludge disintegration. Fig. 3 demonstrates the variation of COD solubilization with specific energy during the chemo-mechanical pretreatment. The specific energy input, defined as the amount of energy that is related to the sludge dry solids content. The rpm is the governing factor for the specific energy input for the rotor of the disperser. The specific energy consumption for the various rpm from 12,000 to 24,000, at different time intervals was monitored with COD solubilization as the response. Specific energy was considered as an important parameter and was varied from 2272 to 54,522 kJ/kg TS. COD solubilization increased steadily as the specific energy input increases, nevertheless, both chemical and mechanical disintegration depended linearly from the specific energy input applied, according to Tiehm et al. (2001). From the figure, it could be distinguished that, the COD solubilization significantly increased at the first stage of energy input range of 4544 kJ/kg TS, thereafter the increase rate slowly saturated. In addition, the COD

solubilization of NaOH at pH 10 and 11 are comparatively high when placed in line with COD solubilization of KOH and Ca(OH)_2 . For NaOH at the lower energy input of 2272 kJ/kg TS, 20–23% of COD solubilization and further increase in energy from 2272 to 54,522 kJ/kg TS, 34–41% of COD solubilization is observed. Similarly, for KOH at the lower energy input, 14–18% and with a further increase in energy 28–35% of COD solubilization is observed. In case of Ca(OH)_2 , at lower energy input 7–9% and with a further increase in energy 16–25% of COD solubilization is only observed. Although the degree of COD solubilization improved by an increase in specific energy input, the improvement was not in direct proportion to the energy input. For example, for mechanical pretreatment, Khanal et al. (2006) obtained SCOD/COD of only 16.2% at an energy input of 66,800 kJ/kg TS. Increasing the specific energy input has not influenced the process to a larger extent other than power consumption. Consequently from the figure it could be noted that, in all the three alkaline conditions at specific energy of 4544 kJ/kg TS, the peak in COD solubilization is noted. The trend of COD solubilization had attained a saturation level, indicating that spending more energy had not caused a significant rise in COD solubilization. Long processing times bring only insignificant improvements in the obtainable fineness. The energy expended serves merely to increase the temperature of the medium (IKA Service Manual, 2007).

Comparing the performance, the COD solubilization% in the case of calcium hydroxide is very low. Some authors have previously reported that Ca(OH)_2 was less efficient than NaOH. Liu et al. (2008) observed the behavior of WAS solubilization using NaOH and Ca(OH)_2 with various alkaline doses. The authors concluded that NaOH was more effective than Ca(OH)_2 for sludge solubilization due to the reason that bivalent cation (Ca^{2+}) is the key matter connecting cell with EPS. Hence, calcium cation helps the dissolved organic polymers to re-flocculate the fragments produced by alkaline treatment. For the similar reason, the NaOH treatment can release more COD than the Ca(OH)_2 treatment when combined with ultrasonic treatment. Further studies with Ca(OH)_2 were not promising and hence only the alkalines NaOH and KOH were selected for subsequent studies. Thus, the rpm and time interval corresponding to specific energy input of 4544 kJ/kg TS, with NaOH and KOH as alkaline agents at pH conditions of 10 and 11 were selected for the second part of biodegradability analysis.

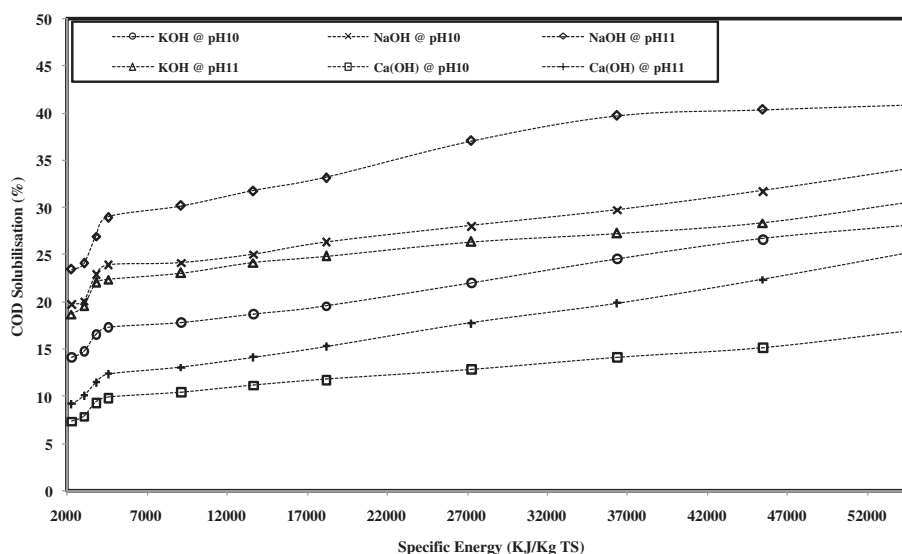


Fig. 3. Variation of COD solubilization with specific energy.

3.1.4. Biochemical methane potential assay

Biodegradability batch assay was carried out to assess the feasibility of using chemo-mechanical pretreatment to improve the anaerobic biological degradation of dairy sludge. Biodegradability assays, wherein cumulative biogas productions were assessed to both raw and pretreated substrates under mesophilic conditions. Cumulative biogas production in serum bottles was maintained for 42 days, and the biogas production was monitored every day. Daily biogas was measured by inserting the needle attached to a syringe (10 and 25 mL). The gas pressure in the bottle displaces the syringe plunger, and the displaced volume was recorded. The results showed that this chemo-mechanical pretreatment of dairy sludge improved anaerobic degradation. The pretreated samples which resulted in better COD solubilization and SS reduction based on specific energy input of 4544 kJ/kg TS (NaOH – pH10; pH 11 and KOH – pH10; pH 11), were chosen as substrates for this BMP assay. Fig. 4 demonstrates the cumulative biogas and methane production with the digestion period at different pretreatment conditions. As seen in figure, none of the digesters showed a lag phase at the beginning of the batch tests. The biogas production rate started increasing potentially since the first day of operation. It is due to the ease of hydrolyzing the compounds and the availability of substrates, which are facilitated by chemo-mechanical pretreatment. The first week of batch digestion is critical because the maximum substrate utilization generally occurs at 5–7 days. Until a digestion time of 6 days, the control and pretreated samples had very similar

biogas yield. The accumulated biogas and methane production at the end of 42 days during the digestion period were nearly 245 and 158 mL, 1005 and 645 mL, 1020 and 654 mL, 920 and 590 mL, 640 and 411 mL for control, NaOH at pH 11, pH 10; KOH at pH 11, pH 10 respectively. Higher yield for NaOH pretreated samples can be explained by the fact that pretreated sludge samples contained higher levels of soluble organic matter which microorganisms were able to use immediately.

The BMP results revealed that, the biogas production for the sludge pretreated at pH 10 and 11 using NaOH were found to be equal and comparable with each other, although NaOH at pH 11 had solubilized a higher fraction of COD and proteins. This suggests that NaOH at pH 11 produce more digestible as well as recalcitrant soluble compounds and in addition, partial biodegradation of solubilized compounds during pretreatment itself might be responsible for lower final biogas volume. The results show that a high COD solubilization from a pretreatment does not necessarily lead to an increased methane production as was seen in Davidsson and Courjansen (2006). On the other hand, pretreatment methods like ultrasonication can lead to increased methane production although the COD solubilization is low as was also seen in Tiehm et al. (2001). In addition, the high concentration of sodium would inhibit the activity of microorganisms and interfere with their metabolism (Chen et al., 2008). These results suggest that with an extended period of 15–42 days, this chemo-mechanical pretreatment enhanced the ultimate degradability of the dairy WAS. Subsequent

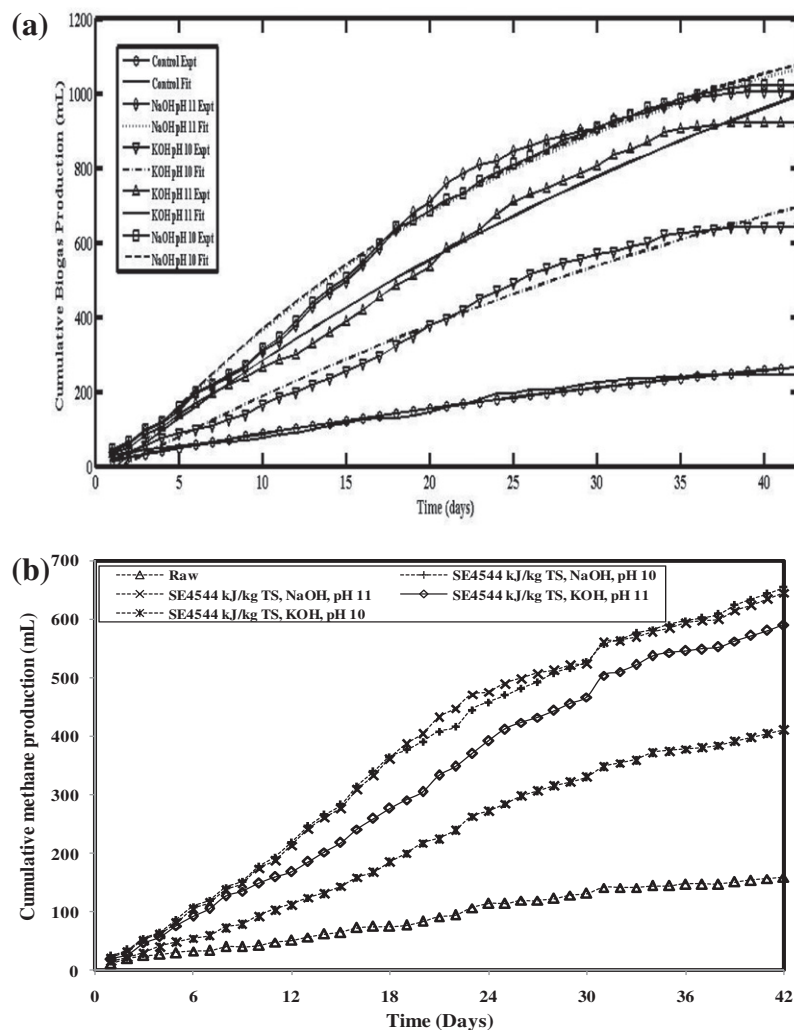


Fig. 4. Cumulative biogas and methane production with digestion period. (a) Cumulative biogas production with model fit. (b) Cumulative methane production.

to the digestion period of 42 days, the biogas production ceased indicating that the digestate did not undergo any further degradation. Percentage of converted COD was calculated based on the Eqs. (5) and (6) and it was found that, NaOH had a COD conversion percentage increased from 21.5% to 82–86%, while for KOH, it was increased from 21.5% to 37–67%. Thus, the COD degradation of the dairy sludge was satisfactorily increased by this pretreatment.

Table 1 presents the kinetic parameters obtained in the optimization process. From the experiment conducted, it can be noted that biogas production is enhanced by an increase from 61% to 76%, higher than control. The trend of biogas production (R_m and P values from Table 1) demonstrates that the solubilization of sludge at NaOH at pH 10 attributes to the significant biogas production. At NaOH at pH 10 the lag time for biogas production is decreased to 0.99 from 1.4. The values were calculated for all experimental conditions and are found to be fit. In the intervening time, this combined treatment led to changes in the physico-chemical characteristics of the sludge. pH and temperature are equally important parameters affecting sludge disintegration efficiency. Fig. 5a demonstrates the variation of pH and temperature with specific energy during the chemo-mechanical pretreatment. This pretreatment had a detrimental effect on the sludge pH. For instance, during chemo-mechanical pretreatment, pH decreased from its initial pH of 10 and 11 to 9.2 and 10.6 for NaOH pretreated sample. This pH drop may be explained by acidic compound formation due to floc disintegration. However, in order to proceed further with anaerobic digestion, the pH had to be neutralized with acids. Higher the resultant pH, higher is the acid's consumption and the treatment costs. Considering the trend in fall of pH at the end of pretreatment, pH 10 is the best initial pH in order to carry out the subsequent anaerobic digestion at a low expense of acid for neutralization. Hence an optimal initial pH of 10 was chosen for further experimental studies. In contrast, temperature increased almost linearly with increasing specific energy. Temperature increased from 28 °C in control to 43 °C for the pretreated sludge with specific energy input of 4544 kJ kg⁻¹ TS. Donsi et al. (2009) reported that the thermal effect resulted from the sludge temperature increase enhanced the sludge disintegration.

This study was attended to find out the effective pretreatment condition with low operating cost, which corresponds to low specific energy input. Consequently, the best performances, in terms of COD solubilization, SS reduction and biogas production, were the ones that occurred for specific energy input of 4544 kJ kg⁻¹ TS for NaOH at pH10, were found to be 24%, 23.3% and 76%, respectively, which might be regarded as a threshold for the predigestion step and these values were adopted as the optimal dose in this combined pretreatment prior to anaerobic digestion tests.

Moreover, it is found that the specific energy consumption of chemo-mechanical treatment decreased (COD solubilization 22%) with the sludge TS increase (25 g/L), showing that it was more efficient for higher TS sludge. An energy balance considering a 76% increase in biogas production and the specific energy requirement of 4544 kJ/kg TS needed in the laboratory device, shows that the process would only be energetically feasible for the higher sludge concentration, as it is a key factor from an economic point of view. Thus, the trade-off between the specific energy requirement and

energy generation from biogas is insignificant. Also, it is extensively known that some sludge disintegration methods are energy demanding, in particular for mechanical and thermal treatment (Zhang et al., 2012). Ultrasonication and microwave irradiation consumed 296% and 627% energy more than this, to achieve comparable sludge disintegration for the similar sludge characteristics (Ahn et al., 2009; Feng et al., 2009). The energy consumption assessment confirmed that this alkaline-disperser treatment was an excellent sludge disintegration method covering a higher range of solids and for the subsequent recovery of biogas.

3.1.5. Soluble protein and carbohydrate release

It is possible that an increase in the COD solubilization that results from the pretreatments might originate from the disruption of microbial cells, causing the release of various organic compounds (e.g. carbohydrates, proteins, lipids and volatile fatty acids). Proteins are the principal constituents of organisms, and they contain carbon, which is a common organic substance as well as hydrogen, oxygen and nitrogen (Nah et al., 2000). For this reason, it was considered that as the level of soluble protein increased, the efficiency of anaerobic digestion would be improved. Due to this pretreatment, solids were solubilized, especially organic solids. Fig. 5b demonstrates the variation on the concentration of soluble proteins and carbohydrates with specific energy. As can be seen, both the soluble protein and carbohydrate concentrations are found to be increased with the specific energy. For example, the soluble protein at a lower specific energy input of 2272 kJ/kg TS was only 525 mg/L, which increased to 589 mg/L at the specific energy of 54522 kJ/kg TS. Similarly, the soluble carbohydrate concentration increased from 119 to 134 mg/L in the specific energy range of 2272–54522 kJ/kg TS. It must be noted that the soluble protein concentration at different specific energies were much higher than those of soluble carbohydrates and the reason for this was attributed to the fact that, protein was the most predominant component and the protein releases presented in the figure are the sums of protein released from EPS as well as the cell lysis. Hence protein concentration is more than carbohydrate. According to Liu and Fang (2002), during the sludge treatment with NaOH, protein is released more compared to carbohydrate, and this result is similar to that obtained in the present study. Furthermore, it was found that increase in energy input doesn't play a major role, i.e. the solubilization rate increases linearly with energy input only till 4544 kJ/kg TS, after which the rise is nearly flat. Thus, specific energy input of 4544 kJ/kg TS is significant for protein and carbohydrate release.

3.1.6. Effect of pretreatment on microscopic image evaluation and microbial enumerations

According to the waste sludge disintegration concept, the particle size of solid particles should be decreased and become more uniform after treatment. In this study, the microscopic view of the dairy WAS samples before and after chemo-mechanical treatment (at optimized condition) was analyzed by the floc images captured from microscopic technology (which are not shown). It was found that, alkaline pretreatment was more efficient in destroying the cell walls and mechanical pretreatment simultaneously emphasized the alkaline effect which was confirmed with the microphotographs. As predictable, the image of the control sample indicated more microbial flocs wrapped with filamentous-like bacteria. The effect of pretreatment on floc structure is apparent and the microscopic observations indicated that, when the sludge was treated with chemo-mechanical treatment for 30 min at 12,000 rpm using NaOH, more or less complete break-up of cell walls was observed with several punctured cells, i.e. the sludge flocs were dispersed, and highly porous flocs were disintegrated into microflocs or flocculi, even primary particles. These

Table 1
Kinetic parameters calculated from the theoretical model for various samples.

Samples	P (mL)	R_m	R^2
Control	482	9.06	0.9954
NaOH at pH 10	1391	51.54	0.9895
NaOH at pH 11	1362	51.50	0.9766
KOH at pH 10	1364	24.24	0.9755
KOH at pH 11	1852	35	0.9743

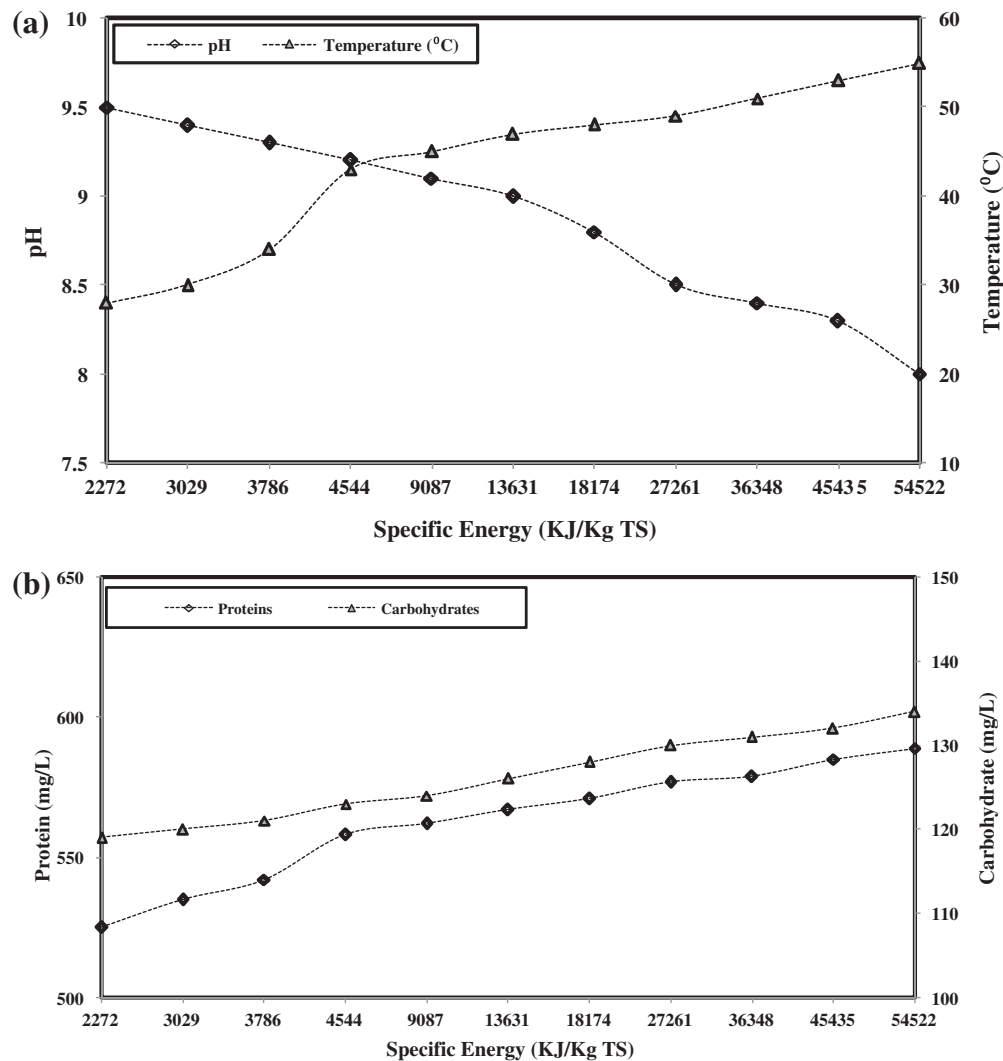


Fig. 5. Variation of pH and temperature; proteins and carbohydrates with specific energy. (a) Variation of pH and temperature. (b) Variation of soluble proteins and carbohydrates.

cavities were caused by the disruption of microbial cells, which resulted in the liberation of intracellular substance to the outside and thus soluble protein concentration increased significantly (Liu et al., 2008). Further it was observed that, there was an obvious change in microbial colony number. The microbial population was effectively decreased in the pretreated sample. The microbial population for pretreated sludge at 10^{-2} dilution was 50 colonies and for the raw sample, 240 colonies at 10^{-4} dilution. pH transformations due to alkali and the mechanical shear stress created by the chemo-mechanical pretreatment process have induced an effective decline in the microbial population. Overall, this new approach not only disintegrates the flocs but simultaneously causes either damage or completely ruptures the cell lysis by giving moderate specific energy input.

4. Conclusions

This study tested the effectiveness of combined alkaline and disperser method for the first time in literature. In this treatment, 4544 kJ kg^{-1} TS of supplied energy for NaOH with pH10 were efficient for sludge disintegration. With the optimal condition, COD solubilization and SS reduction can be improved by 24% and 23.3% higher than that of control. BMP assay results of pretreated

sludge confirmed that the observed solubilization led to an increase in sludge biodegradability, nearly 76% higher biogas production than control. These experimental results indicated that this combination pretreatment was useful for the enhancement of biodegradability.

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